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VUA-SEN307 Database Systems — Week 4 Practical

Lab 2: Database Design and Normalization

ENTITIES AND ATTRIBUTES

ENTITY	ATTRIBUTES	PRIMARY KEY
Department	DepartmentName	DepartmentName
Student	MatricNo, name	MatricNo
Lecturer	name	name
Course	courseCode, courseTitle	courseCode
Enrolment	session	Composite key: MatricNo + courseCode + session

UNNORMALIZED REGISTRATION SHEET

The original registration sheet contains multiple course codes, course titles and lecturers in some cells. The records are expanded into individual student-course rows in 1NF below.

1NF: First Normal Form

Lists were split and individual rows were created. Each row represents one student-course registration.

Matric No	Student Name	Student Dept	Course Code	Course Title	Lecturer	Lecturer Dept	Session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Engineering	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Engineering	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Engineering	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Science	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Engineering	2026/2027

Primary key: (Matric No, Course Code, Session). This composite key uniquely identifies each student-course registration within a session.

2NF: Second Normal Form

Student details were separated from course details and enrolment details to remove partial dependencies.

Table 1: Student

Matric No (PK)	Student Name	Student Dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Table 2: Course

Course Code (PK)	Course Title	Lecturer	Lecturer Dept
SEN307	Database Systems	Dr A. Okafor	Software Engineering
SEN301	Software Architecture	Dr M. Bello	Software Engineering
COS301	Operating Systems	Mrs F. Eze	Computer Science

Table 3: Enrolment

Matric No (PK, FK)	Course Code (PK, FK)	Session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027

Matric No (PK, FK)	Course Code (PK, FK)	Session
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

2NF explanation: Student attributes depend on Matric No, course attributes depend on Course Code, and each enrolment is identified by the student, course and session.

3NF: Third Normal Form

Department and lecturer details were separated into their own tables, and foreign keys link related records.

Table 1: Department

Dept ID (PK)	Dept Name
1	Software Engineering
2	Computer Science

Table 2: Student

Matric No (PK)	Name	Dept ID (FK)
20231234	Ada Obi	1
20231235	John Musa	2
20231236	Grace Okon	1

Table 3: Course

Course Code (PK)	Course Title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Table 4: Lecturer

Lecturer ID (PK)	Lecturer Name	Dept ID (FK)
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Table 5: Enrolment

Matric No (PK, FK)	Course Code (PK, FK)	Session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

Table 6: CourseAndLecturer

Course Code (PK, FK)	Lecturer ID (PK, FK)	Session (PK)
SEN307	1	2026/2027
SEN301	2	2026/2027
COS301	3	2026/2027

FINAL SCHEMA

1. Department (dept_id PK, dept_name)

2. Student (matric_no PK, name, dept_id FK → Department)

3. Lecturer (lecturer_id PK, lecturer_name, dept_id FK → Department)

4. Course (course_code PK, course_title)

5. Enrolment (matric_no PK/FK → Student; course_code PK/FK → Course; session)

6. CourseAndLecturer (course_code PK/FK → Course; lecturer_id PK/FK → Lecturer; session PK)

Conclusion: The registration database was organized into related tables using primary keys, foreign keys and normalization principles to reduce duplicated data and improve consistency.